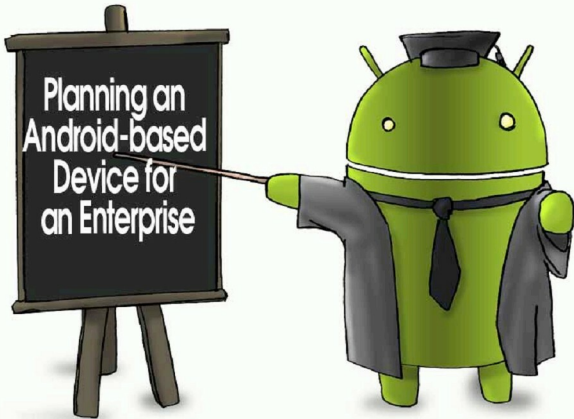




Android's favourable licence, established application ecosystem and well-recognised user experience has made it the preferred platform for any modern embedded system. The pluggable architecture enables integration with a variety of desirable software and hardware components with low coupling. At the same time, support for a range of hardware platforms enables the designing of cost-effective low-end devices like digital watches, as well as sophisticated high-end devices like in-vehicle infotainment systems and home automation systems. This article covers the crucial steps of analysing an enterprise's needs, understanding SoCs, and deciding which device addresses the need while fitting the budget.

The first step in creating a custom device is to have a clear picture of what the device is required to do, and what the company's budget is. Depending on these two factors, the exact hardware requirements can be figured out. For instance, if the device is required for the stock management staff of a grocery centre, then the bar-code scanning capability would be required and it may be appropriate to choose a small form-factor smartphone for the task, with Android version 2.2 or above. If the device is to be designed for sales people to upload customer data on the go, then Wi-Fi connectivity would be desirable, and a tablet form-factor with Android 2.3 or above would work well for such a requirement.

In order to understand the complete workflow of device creation, let us consider the following target system as a sample scenario:

- The AGROSTAR company deals in a wide range of agriculture products covering a range of requirements from crop protection to animal health. AGROSTAR

requires a custom device solution for its field sales team that visits client sites at various locations in India.

- The company has a budget limit of Rs 5000 per device, and would need the solution deployed for 3000+ field workers.
- The primary use cases are shown in the following diagram on the next page.

Looking at the above system requirements and the users' need to show presentations and product details to prospective customers, a tablet form factor, with Wi-Fi capability and Android 2.3+ seems suitable for the target requirement.

Identify device hardware

Once the system requirements are identified, the second step is to decide on hardware requirements. Though creating a custom device may be useful for very specific use cases and bulk production, enterprises prefer successful devices that are already in use in the market. So let us learn more about the device hardware and the hardware profiles used by popular